

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1 loading packages and checking working directory
#install.packages("tidyverse")
#install.packages("rvest")
#install.packages("ggplot2")
#install.packages("here")
library(tidyverse)
library(rvest)
library(ggplot2)
library(here)

getwd()
```

```
## [1] "/home/guest/ENV 872"
```

```
here()
```

```
## [1] "/home/guest/ENV 872"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2 reading in the scraping html
```

```
webpage <- read_html(
  'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')
webpage
```

```
## {html_document}
## <html xmlns="http://www.w3.org/1999/xhtml" lang="en" xml:lang="en">
## [1] <head>\n<title>DWR :: Local Water Supply Planning</title>\n<meta http-equ ...
## [2] <body id="plan">\r\n<!--<div id="division-header">\r\n<a name="top" href= ...
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3 scraping website values
```

```
water_system_name <- webpage %>%
  html_nodes('div+ table tr:nth-child(1) td:nth-child(2)') %>%
  html_text()
water_system_name
```

```
## [1] "Durham"
```

```
PWSID <- webpage %>%
  html_nodes('td tr:nth-child(1) td:nth-child(5)') %>%
  html_text()
PWSID
```

```
## [1] "03-32-010"
```

```
ownership <- webpage %>%
  html_nodes('div+ table tr:nth-child(2) td:nth-child(4)') %>%
  html_text()
ownership
```

```
## [1] "Municipality"
```

```
MGD <- webpage %>%
  html_nodes('th~ td+ td , th~ td+ td') %>%
  html_text()
MGD
```

```
## [1] "34.5000" "36.0600" "37.3300" "32.1000" "46.6500" "37.3600" "38.2000"
## [8] "41.9000" "36.5800" "36.7300" "42.9600" "34.4500"
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in date format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

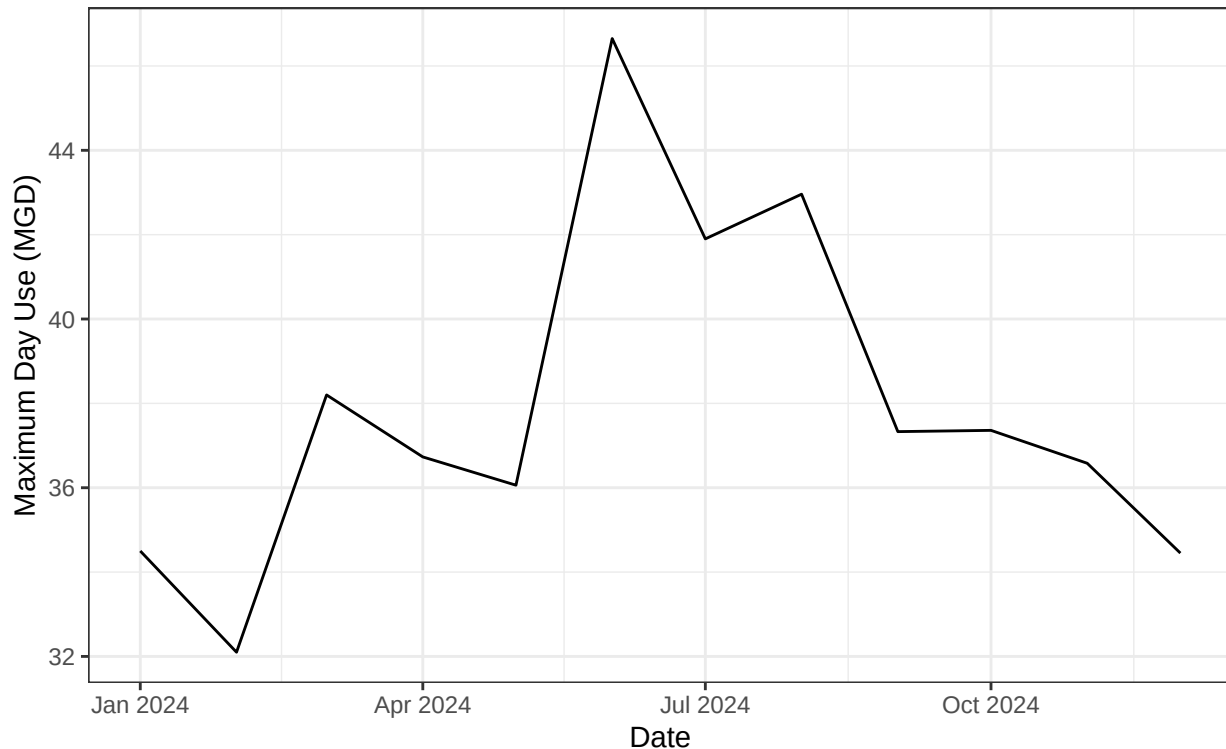
5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```
#4 creating data frame
year <- '2024'
water_data <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
  "Water system name" = water_system_name,
  "PWSID" = PWSID,
  "Ownership" = ownership,
  "Maximum Day Use (MGD)" = as.numeric(MGD)) %>%
  mutate(Date = my(paste(Month, "-", year)))

#5 line plot
ggplot(water_data, aes(Date, Maximum.Day.Use..MGD.)) + geom_line() +
  ggtitle('Monthly maximum daily water use') +
  labs(subtitle = "Durham Municipality, 2024", y = "Maximum Day Use (MGD)") +
  theme_bw()
```

Monthly maximum daily water use

Durham Municipality, 2024



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6. creating function to scrape website
scrape_website <- function(PWSID, year){
  website <- read_html(paste0(
    'https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=',
    PWSID, '&year=', year))

  water_system_name_tag <- 'div+ table tr:nth-child(1) td:nth-child(2)'
  PWSID_tag <- 'td tr:nth-child(1) td:nth-child(5)'
  ownership_tag <- 'div+ table tr:nth-child(2) td:nth-child(4)'
  MGD_tag <- 'th~ td+ td , th~ td+ td'

  water_system_name <- website %>% html_nodes(water_system_name_tag) %>%
    html_text()
  PWSID <- website %>% html_nodes(PWSID_tag) %>% html_text()
  ownership <- website %>% html_nodes(ownership_tag) %>% html_text()
  MGD <- website %>% html_nodes(MGD_tag) %>% html_text()
```

```

function_data <- data.frame("Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
                             "Year" = rep(year,12),
                             "Water system name" = water_system_name,
                             "PWSID" = PWSID,
                             "Ownership" = ownership,
                             "Maximum Day Use (MGD)" = as.numeric(MGD)) %>%
mutate(Date = my(paste(Month, "-", year)))

return(function_data)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7 scraping data for 2020, plotting it
data_2020 <- scrape_website('03-32-010', 2020)

glimpse(data_2020)

```

```

## Rows: 12
## Columns: 7
## $ Month                <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Year                  <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, ~
## $ Water.system.name    <chr> "Durham", "Durham", "Durham", "Durham", "Durham"~
## $ PWSID                 <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-01~
## $ Ownership            <chr> "Municipality", "Municipality", "Municipality", ~
## $ Maximum.Day.Use..MGD. <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29,~
## $ Date                  <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02-01, ~

```

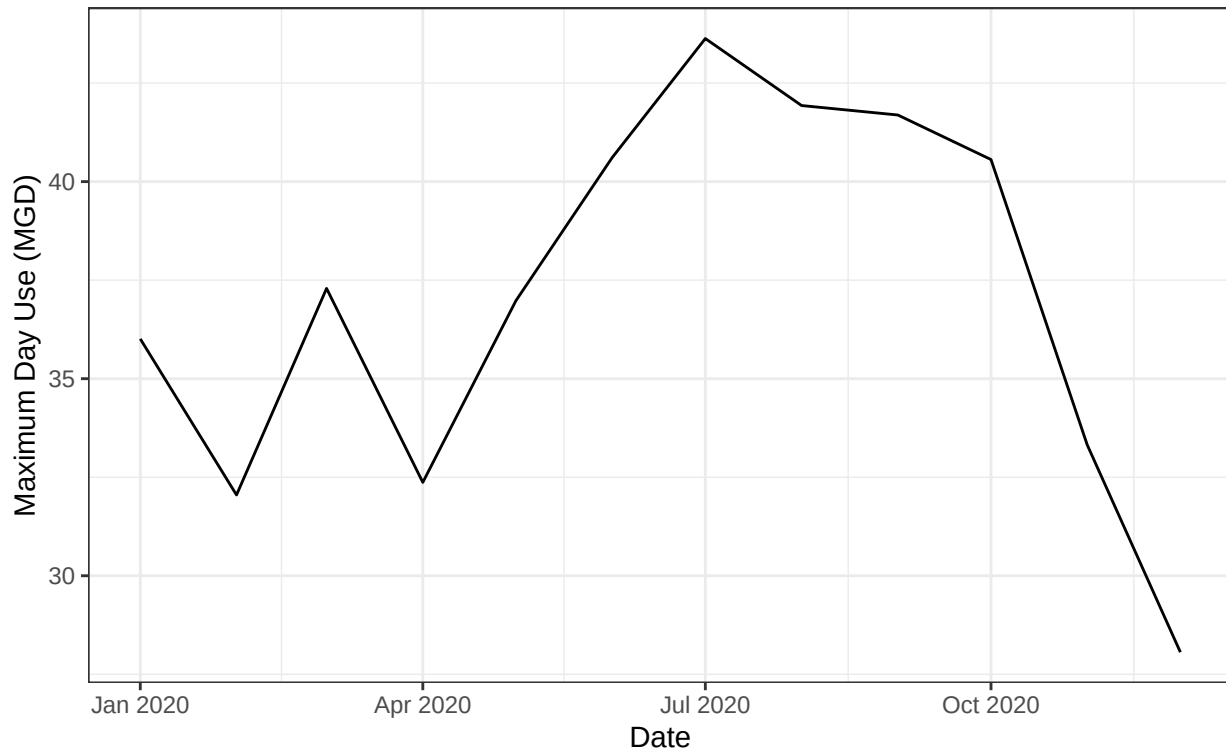
```

ggplot(data_2020, aes(Date, Maximum.Day.Use..MGD.)) + geom_line() +
  ggtitle('Monthly maximum daily water use') +
  labs(subtitle = "Durham Municipality, 2020", y = "Maximum Day Use (MGD)") +
  theme_bw()

```

Monthly maximum daily water use

Durham Municipality, 2020



8. Use the function above to extract data for Cary (PWSID = '03-92-020') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

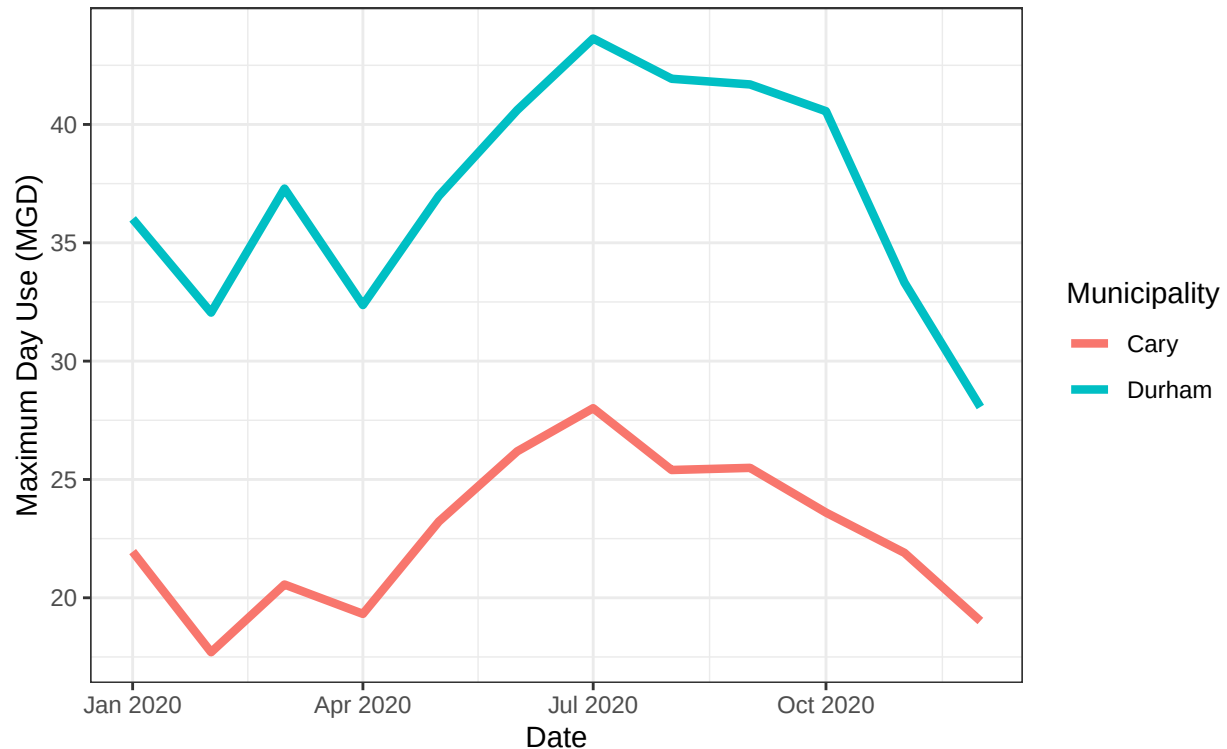
#8 scraping data from Cary for 2020 and combining with Durham 2020

```
cary_2020 <- scrape_website('03-92-020', 2020)
all_2020 <- rbind(data_2020, cary_2020)

ggplot(all_2020, aes(Date, Maximum.Day.Use..MGD., colour = Water.system.name)) +
  geom_line(linewidth = 1.5) +
  ggtitle('Monthly maximum daily water use') +
  labs(subtitle = "Durham and Cary, 2020", y = "Maximum Day Use (MGD)",
        color = "Municipality") +
  theme_bw()
```

Monthly maximum daily water use

Durham and Cary, 2020



9. Use the code & function you created above to plot Cary's max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = 'loess').

TIP: See Section 3.2 in the "10_Data_Scraping.Rmd" where we apply "map2()" to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

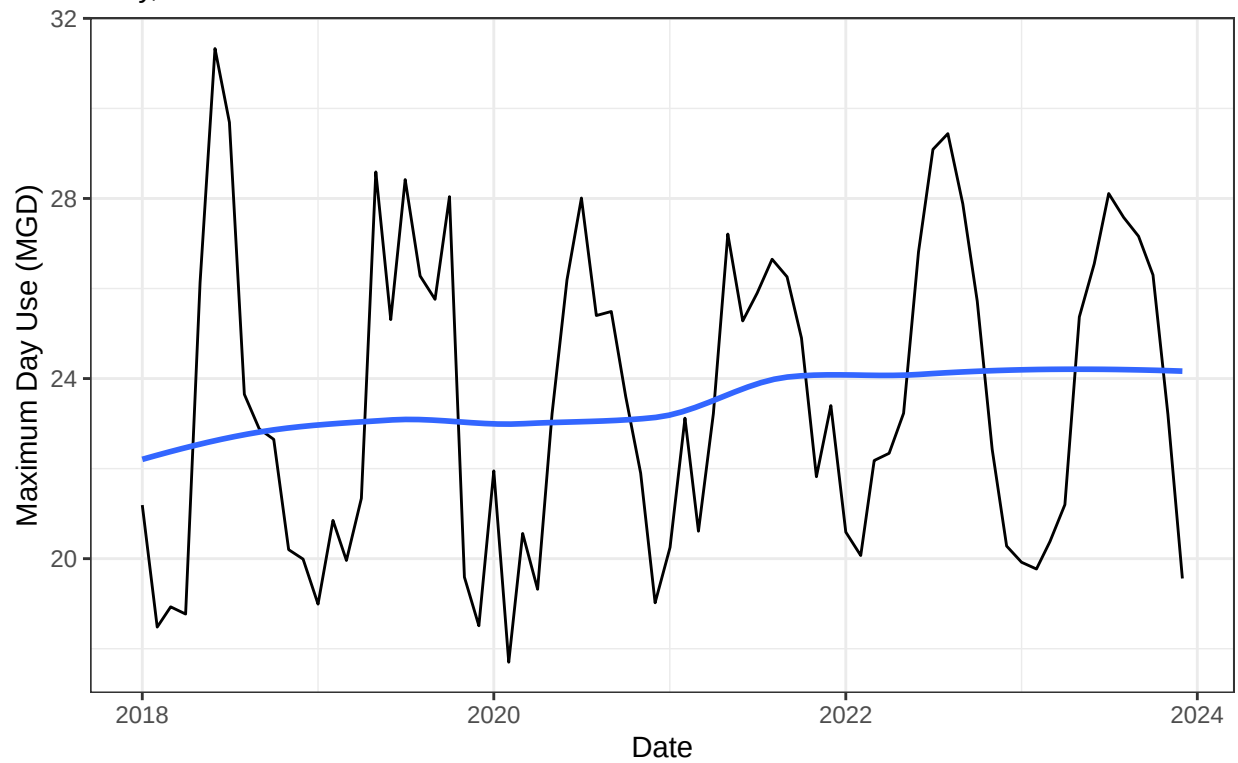
```
#9 Cary data from 2018 to 2023
the_years <- c(2018:2023)
all_Cary <- map2('03-92-020', the_years, scrape_website) %>%
  bind_rows()

ggplot(all_Cary, aes(Date, Maximum.Day.Use..MGD.)) + geom_line() +
  geom_smooth(method = "loess", se = F) +
  ggtitle('Monthly maximum daily water use') +
  labs(subtitle = "Cary, 2018-2023", y = "Maximum Day Use (MGD)") +
  theme_bw()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Monthly maximum daily water use

Cary, 2018–2023



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time?

Answer: Cary is showing a small trend overtime of increasing water usage from 2018 to 2023.